



**Technical School of
Telecommunications Engineering
and Systems**



ANTENNAS AND EMC

SOLUTION

**LABORATORY PRACTICE 1
WIRE ANTENNA ANALYSIS**

Last Name and First Name:

Aladro Abad, Guillermo

González García, Rodrigo



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1. Introduction

This report presents the work and results of Laboratory Practice 1, devoted to the analysis of wire antennas using the method of moments with the FEKO electromagnetic simulation software. The practice has two objectives: to reinforce the fundamental concepts of the course (input impedance, current distribution, radiation patterns, field regions, and bandwidth) and to develop proficiency in the use of the simulation tool.

The document follows the order of the laboratory instructions. First, the influence of mesh size on the convergence of the results is studied (Section 2). Next, the radiation regions of a dipole are analyzed from the spatial variation of its near fields (Section 3). Finally, several basic wire antennas are designed and characterized: a dipole, a monopole, reduced-height monopoles (capacitive loading, an inverted-L antenna, and an IFA), and loop antennas (Section 4).

All calculations were performed at the operating frequency derived from the ID number (02757023): $f = 100 \cdot C3 + 10 \cdot C2 + C1 = 100 \cdot 0 + 10 \cdot 2 + 3 = 23$ MHz, corresponding to a free-space wavelength $\lambda = c/f = 13,043$ m. The wire diameter is $d = \lambda / (150 + 10 \cdot C5 + C4) = \lambda / 207 = 63,0$ mm. The reference standing-wave ratios used for the bandwidth calculations are $SWR = 1,5 + 0,1 \cdot C7 = 1,7$ for the dipole in Section 4.1 ($C7 = 2$), and $SWR = 1,5 + 0,1 \cdot C8 = 1,5$ for the remaining antennas ($C8 = 0$).

2. Mesh Type to Use

The five meshes specified in the laboratory instructions are compared:

- Custom, segment length $\lambda/4$
- Coarse
- Standard
- Fine
- Custom, segment length $\lambda/50$

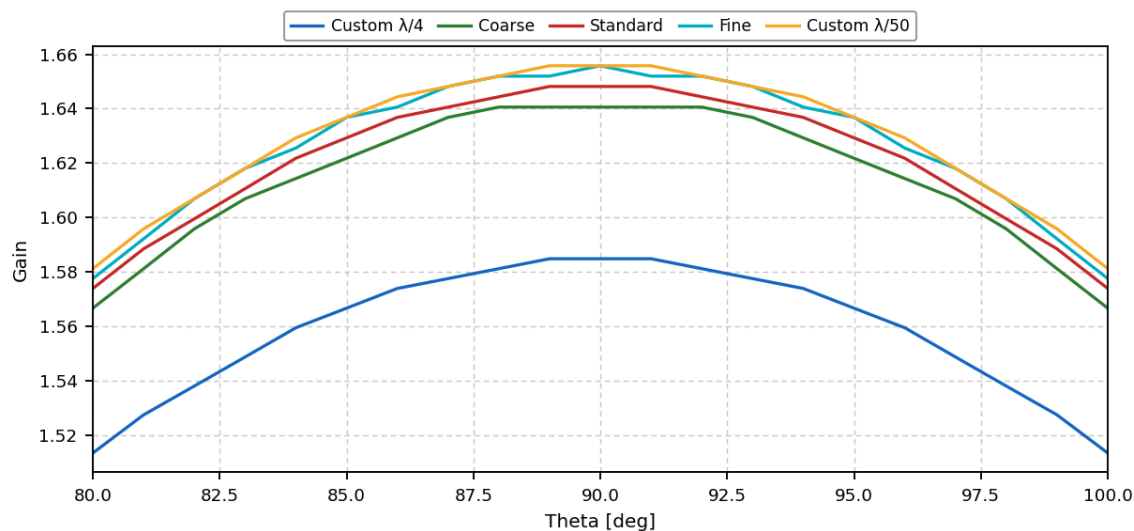
2.1. Prob1A

A dipole of length $\lambda/2 = 6,522$ m is implemented along the z axis, center-fed by a 1 V voltage source, and solved using the five meshes. The table lists the input impedance and the number of segments into which the wire is divided.

Mesh	Custom $\lambda/4$	Coarse	Standard	Fine	Custom $\lambda/50$
Impedance (Ω)	8.3181E+01 + j4.5989E+01	8.6272E+01 + j4.8656E+01	8.7222E+01 + j4.9208E+01	8.8954E+01 + j5.0117E+01	9.0407E+01 + j5.0733E+01
Number of segments	3	7	9	15	25

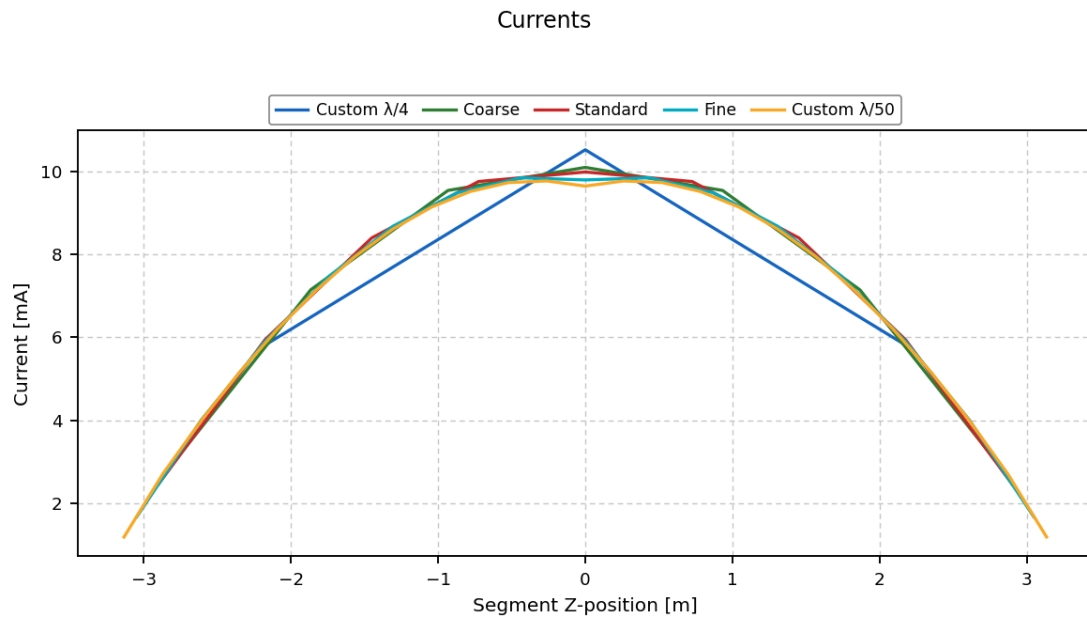
→ Far field:

Far field



Total Gain (Frequency = 23 MHz; Phi = 0 deg) - Prob1A

→ **Current magnitude:**



The results converge monotonically as the mesh is refined. Between the coarsest mesh ($\lambda/4$, 3 segments) and the finest mesh ($\lambda/50$, 25 segments), the input resistance increases from $83,2 \Omega$ to $90,4 \Omega$ ($\approx 8 \%$) and the reactance from $46,0 \Omega$ to $50,7 \Omega$, whereas the difference between Fine and $\lambda/50$ is already below 2% . The same behavior is observed in the radiated field for the $\phi = 0^\circ$ cut: the $\lambda/4$ mesh underestimates the gain (1,58 versus 1,66 in linear units at the maximum, $\theta = 90^\circ$). The current distribution shows that, with 3 segments, the half-sine profile is reduced to a triangular shape and the feed current is overestimated (10,5 mA versus 9,8 mA with $\lambda/50$), as a direct consequence of the lower calculated impedance. With segment lengths on the order of $\lambda/20$ - $\lambda/50$, the solution can be considered converged, which supports the practical rule of meshing wires with segments no longer than $\lambda/10$ - $\lambda/20$.

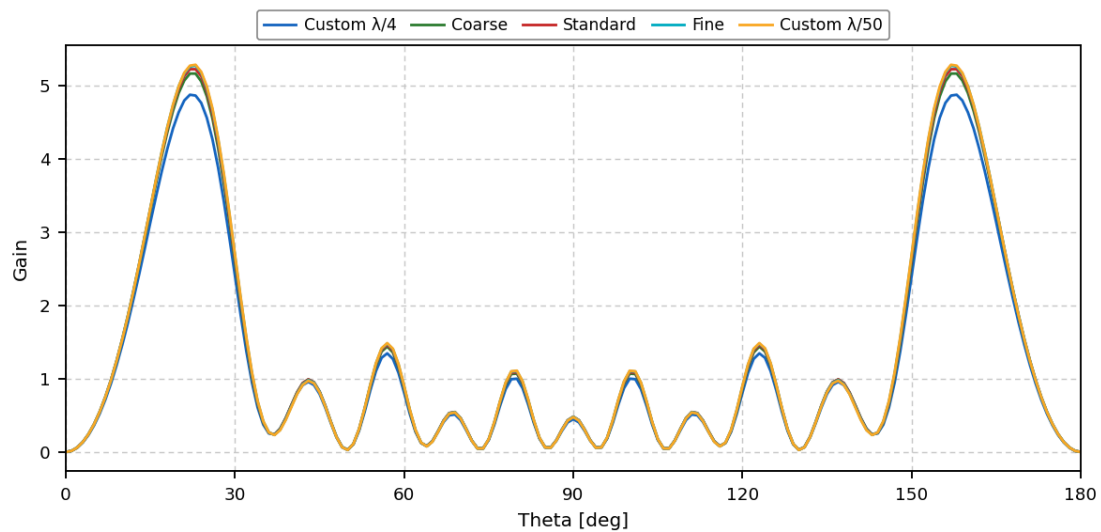
2.2. Prob1B

The same study is repeated for a dipole of length $11\lambda/2 = 5,5\lambda = 71,74 \text{ m}$.

Mesh	Custom $\lambda/4$	Coarse	Standard	Fine	Custom $\lambda/50$
Impedance (Ω)	$1.5900\text{E}+02 + \text{j}6.8187\text{E}+01$	$1.6130\text{E}+02 + \text{j}6.1050\text{E}+01$	$1.6392\text{E}+02 + \text{j}5.8724\text{E}+01$	$1.6851\text{E}+02 + \text{j}5.5765\text{E}+01$	$1.7196\text{E}+02 + \text{j}5.3569\text{E}+01$
Number of segments	25	47	69	141	275

→ Far field:

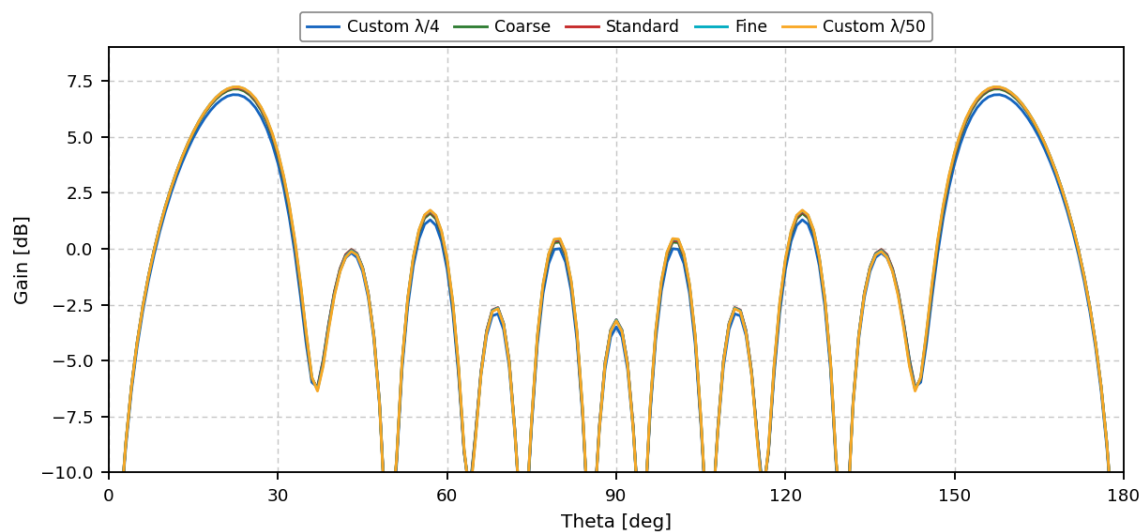
Far field



Total Gain (Frequency = 23 MHz; Phi = 0 deg) - Prob1B

→ Far field in dB:

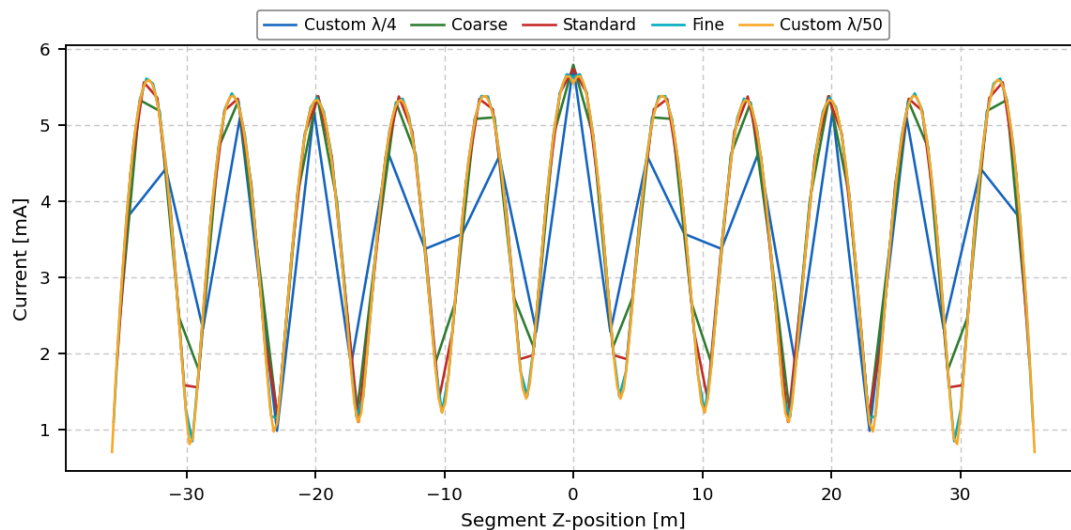
Far field



Total Gain [dB] (Frequency = 23 MHz; Phi = 0 deg) - Prob1B

→ **Current magnitude:**

Currents



Current Magnitude (Frequency = 23 MHz) - Prob1B

For an electrically long antenna, the effect of the mesh is much more evident. With $\lambda/4$ segments there are barely two samples per half-cycle of the standing current wave, so the eleven current lobes are poorly represented and the radiation pattern loses level in the main lobes (6,88 dBi versus 7,23 dBi with $\lambda/50$, with the maximum at $\theta \approx 22^\circ$ - 23°). The impedance magnitude varies by about 8 % between the two extreme meshes and, again, the difference between Fine and $\lambda/50$ is small (≈ 2 % in the real part). In accordance with the laboratory instructions, all simulations from this point onward use the Fine mesh.

3. Antenna Radiation Regions

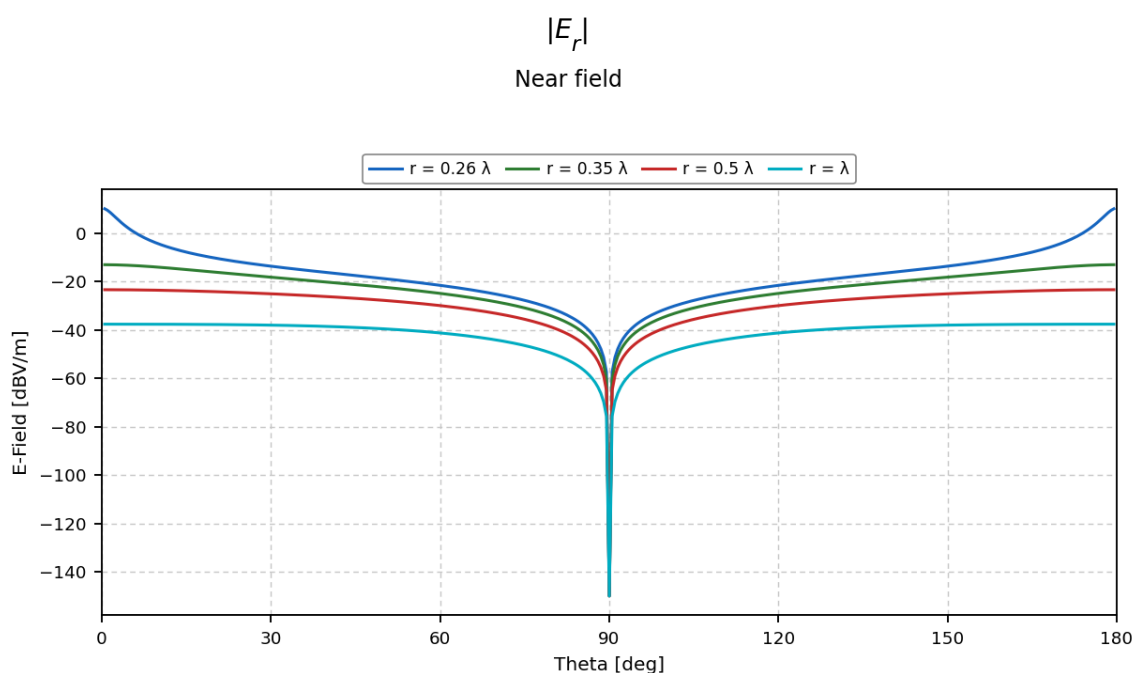
The fields around an antenna contain terms that decay as $1/r$ (radiation terms) and terms proportional to $1/r^2$ and $1/r^3$ (reactive or induction terms). For a z-directed dipole analyzed in the $\phi = 0^\circ$ plane, this means:

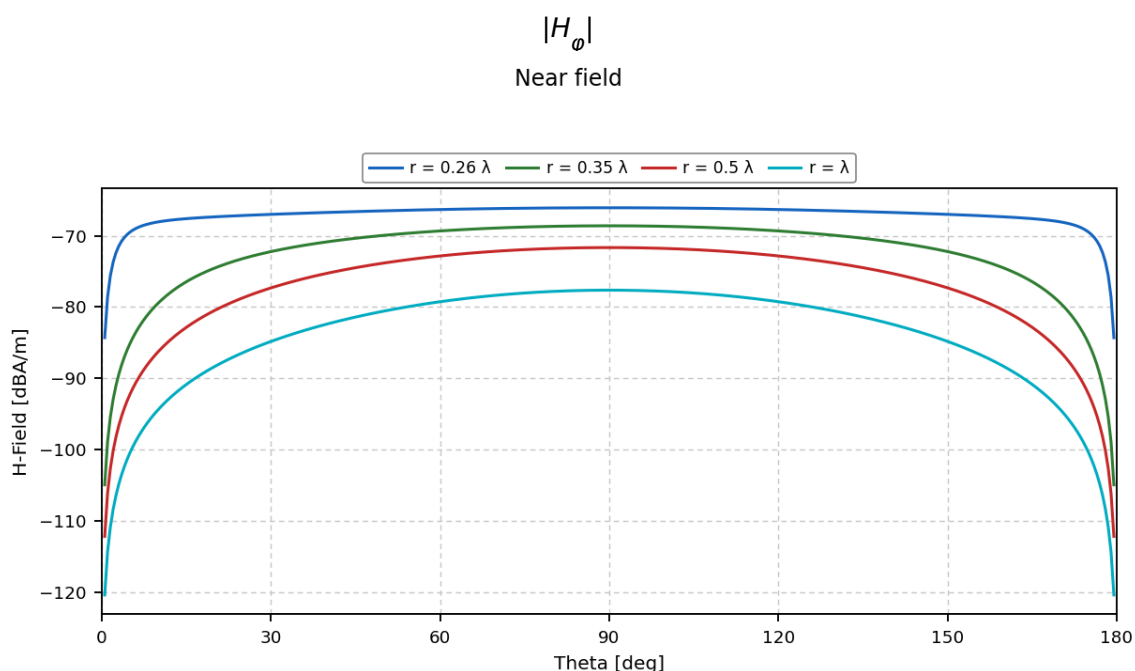
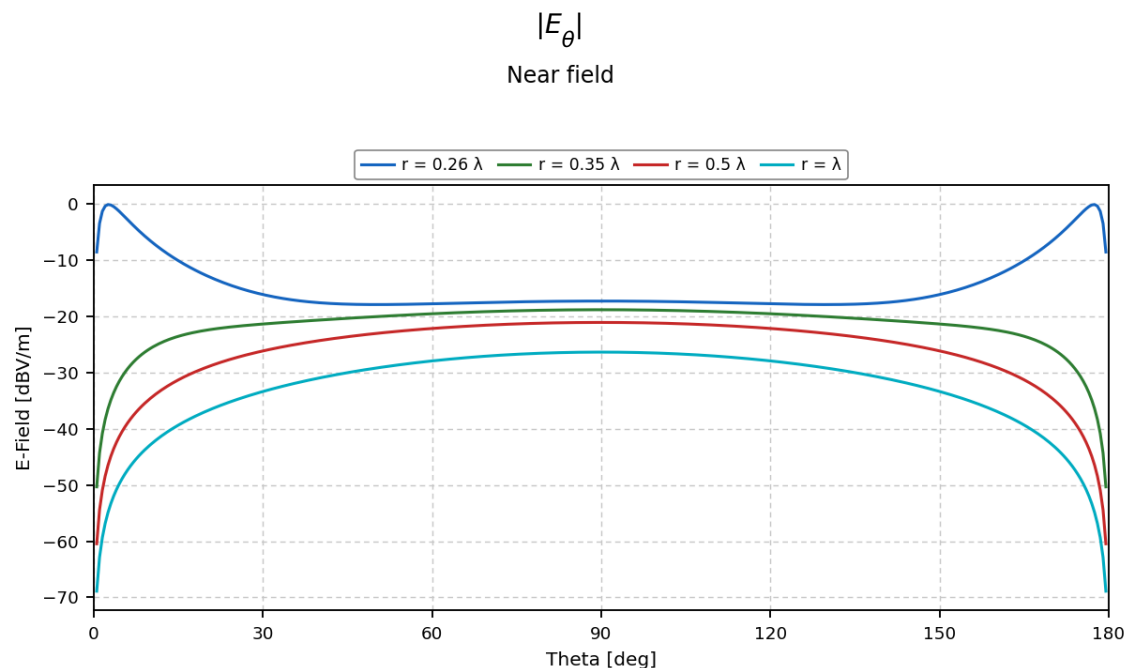
- Radial component E_r : it contains only $1/r^2$ and $1/r^3$ terms; it represents stored reactive energy and must vanish as the observation point moves away from the antenna.
- Transverse components E_θ and H_ϕ : they contain the $1/r$ radiation term and carry power into the far field, where their ratio approaches the free-space wave impedance $\eta = 377 \Omega$.
- Components E_ϕ , H_r , and H_θ : they are identically zero because of the rotational symmetry of the dipole; as stated in the laboratory instructions, it is sufficient to indicate this and they are not plotted.

The usual expressions are used as the boundaries between regions: the reactive region extends to $r = 0,62 \cdot \sqrt{(D^3/\lambda)}$, the Fresnel region (radiating near field) lies between that limit and $r = 2D^2/\lambda$, and the Fraunhofer region or far field begins beyond that point, where D is the maximum dimension of the antenna.

3.1. Prob2A $\rightarrow \lambda/2$ Dipole

For the $\lambda/2$ dipole ($D = 0,5\lambda$), the preceding boundaries are $0,22\lambda$ and $0,5\lambda$. Therefore, $r = 0,26\lambda = 3,39$ m and $r = 0,35\lambda = 4,57$ m lie in the Fresnel region, very close to the boundary of the reactive region; $r = 0,5\lambda = 6,52$ m coincides with the $2D^2/\lambda$ boundary; and $r = \lambda = 13,04$ m is already in the far field. The magnitudes of the nonzero E- and H-field components are plotted below as functions of θ for the four radii.





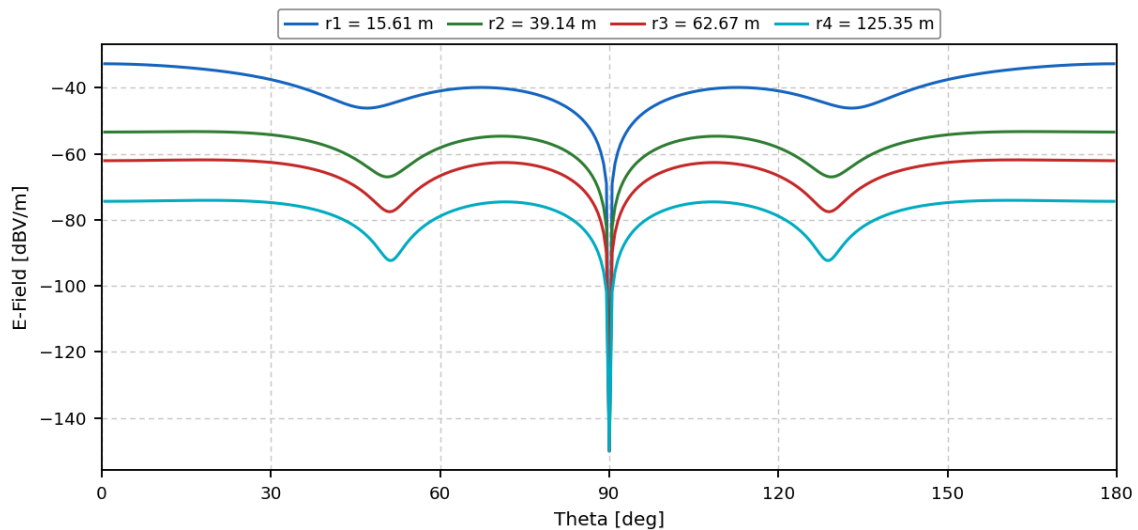
The components E_{ϕ} , H_r , and H_{θ} are zero, as required by the azimuthal symmetry of the antenna.

The plots confirm the expected behavior. $|E_r|$ has a null at $\theta = 90^\circ$ (there is no radial component in the equatorial plane because of the symmetry of the current distribution) and increases toward the axis, where the reactive character dominates. From $r = 0,26\lambda$ to $r = \lambda$ it decreases by 21,5 dB (measured at $\theta = 45^\circ$), clearly faster than $1/r$. In contrast, $|E_{\theta}|$ and $|H_{\phi}|$ decrease by only ≈ 12 dB over the same distances, exactly as predicted by the $1/r$ law ($20 \cdot \log(13,04/3,39) = 11,7$ dB), and retain their half-sine angular shape at all distances. In addition, at $r = \lambda$ the ratio $|E_{\theta}|/|H_{\phi}|$ at $\theta = 90^\circ$ is 365Ω , already very close to $\eta = 377 \Omega$: the wave is locally plane and the observation point is in the far field.

3.2. Prob2B $\rightarrow$ 1,55 λ Dipole

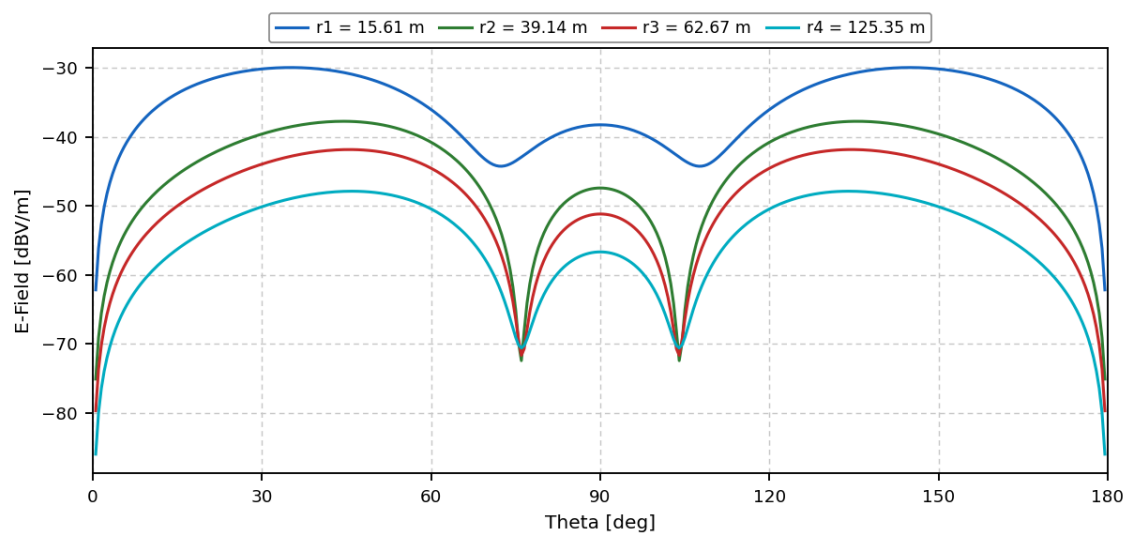
The second dipole has a length $(1,2 + 0,05 \cdot C6) \cdot \lambda = 1,55\lambda = 20,22 \text{ m}$ ($C6 = 7$). For $D = 1,55\lambda$, the Fresnel region is bounded by $r1 = 0,62 \cdot \sqrt{(D3/\lambda)} = 1,196\lambda = 15,61 \text{ m}$ at its beginning and $r3 = 2D2/\lambda = 4,805\lambda = 62,67 \text{ m}$ at its end. The fields are calculated at $r1$, at the midpoint of the region ($r2 = 39,14 \text{ m}$), at $r3$, and at twice the far-field boundary ($r4 = 125,35 \text{ m}$). Because $r1$ is comparable to the half-arm length of the antenna ($10,11 \text{ m}$), the fields are not evaluated exactly at $\theta = 0^\circ$ and 180° .

$|E_r|$
Near field



R E-Field Magnitude (Frequency = 23 MHz; Phi = 0 deg) - Prob2B

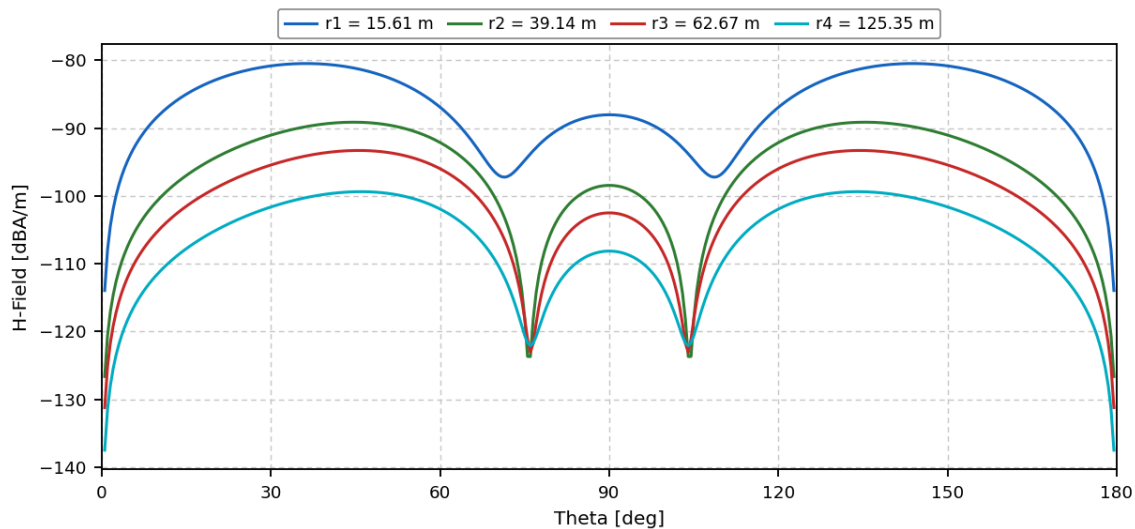
$|E_\theta|$
Near field



Theta E-Field Magnitude (Frequency = 23 MHz; Phi = 0 deg) - Prob2B

$$|H_{\phi}|$$

Near field



Phi H-Field Magnitude (Frequency = 23 MHz; Phi = 0 deg) - Prob2B

Again, E_{ϕ} , H_r , and H_{θ} are zero by symmetry. The radial component decreases by 36,3 dB between r_1 and r_4 ($\theta = 45^\circ$), compared with the 18,1 dB expected from a $1/r$ law for this distance ratio ($\times 8$); it follows an almost exact $1/r^2$ dependence, the dominant induction term. $|E_{\theta}|$ and $|H_{\phi}|$ decrease by ≈ 17 dB (the $1/r$ law), and their angular shape evolves with distance: at r_1 the lobes are distorted by the reactive terms and by path-length differences from the various points of the antenna; at r_2 some ripple is still visible; and from r_3 onward the angular shape no longer changes (its level only decreases as $1/r$), with lateral maxima at $\theta \approx 40^\circ$ and 140° and a smaller equatorial lobe, as is typical of a $1,55\lambda$ dipole. The antenna has reached its far-field pattern: at r_4 the ratio $|E_{\theta}|/|H_{\phi}|$ is $375 \Omega \approx \eta$.

4. Basic Antenna Design

4.1. $\lambda/2$ Dipole

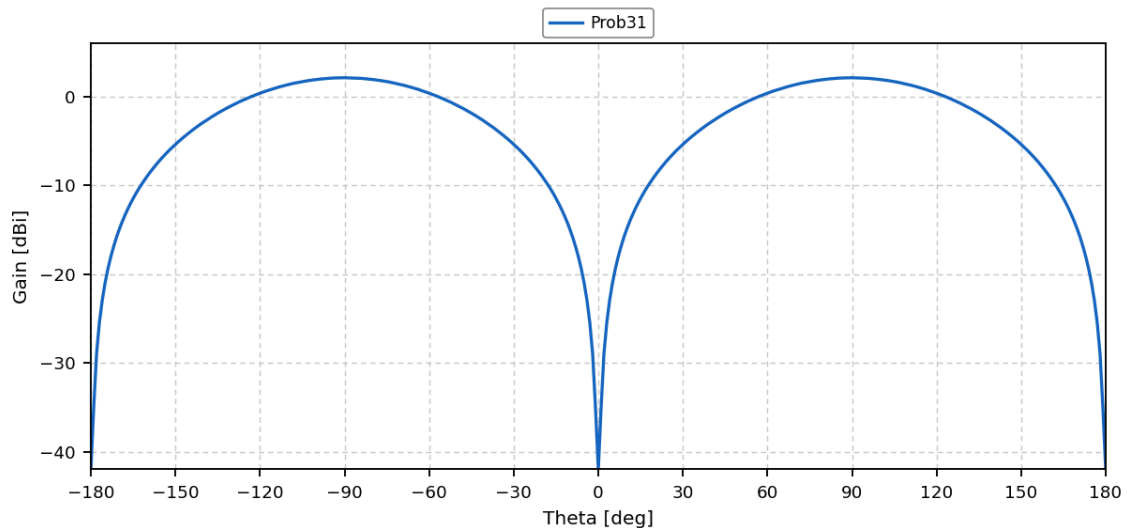
The objective is to find the dipole length closest to $\lambda/2$ for which the input impedance is real. Starting from $L = \lambda/2$ (which has a reactance of $+50 \Omega$, according to Section 2.1), the length is reduced until the imaginary part of the impedance is zero. The resulting design is summarized below:

- Resonant length: $L = 6,0961 \text{ m} = 0,4674\lambda$ (6,5 % shorter than $\lambda/2$; the shortening is consistent with the wire thickness, $d = \lambda/207$).
- Input impedance: $7.1783\text{E}+01 - j4.1477\text{E}-07 \Omega$.
- Gain: 2,13 dBi (at $\theta = 90^\circ$).

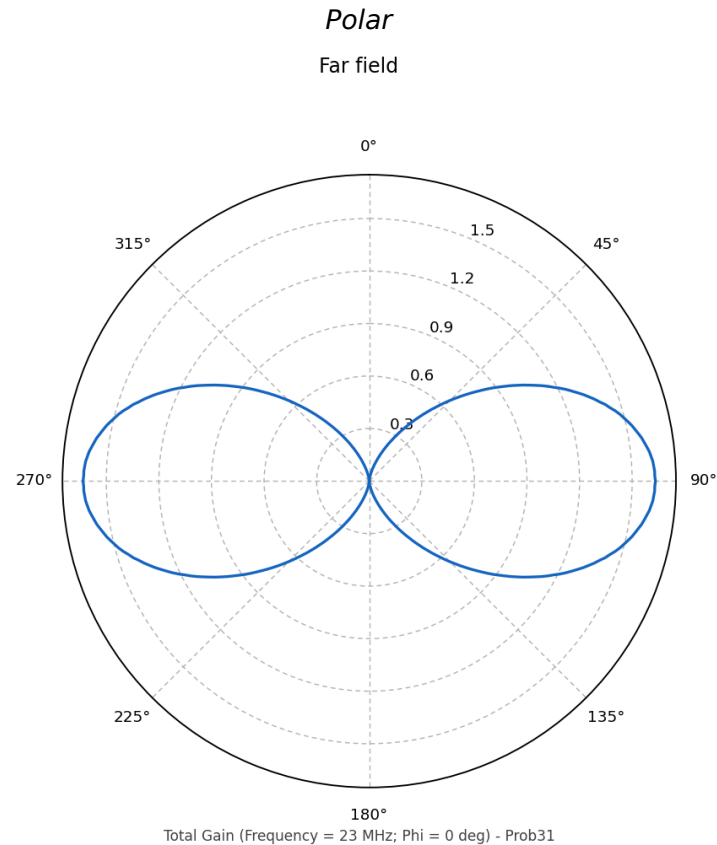
→ **Radiation pattern:**

Cartesian

Far field



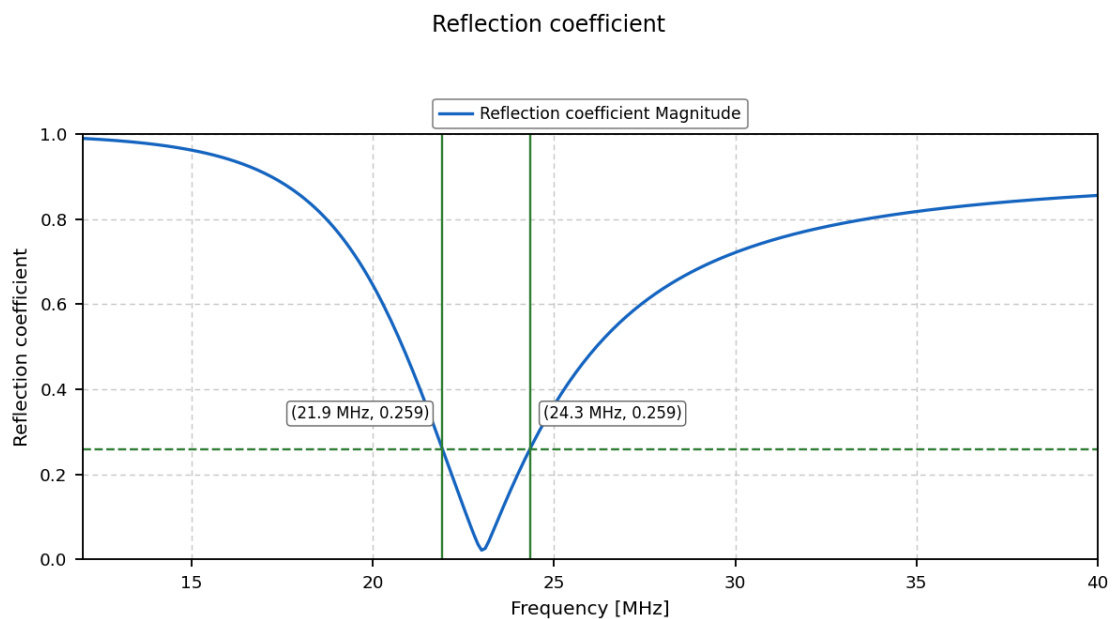
Total Gain [dBi] (Frequency = 23 MHz; Phi = 0 deg) - Prob31



- The -3 dB beamwidth of the designed antenna is $129,2^\circ - 50,8^\circ = 78,4^\circ$.

Determine the bandwidth of this dipole if the standing-wave ratio must be below 1,7, using 75 Ω as the reference impedance. State how the antenna gain varies over this bandwidth.

SWR = 1,7 corresponds to $|\Gamma| = (SWR - 1)/(SWR + 1) = 0,7/2,7 = 0,2593$. In the reflection-coefficient plot referenced to 75 Ω , the frequencies at which $|\Gamma|$ crosses this value are identified:

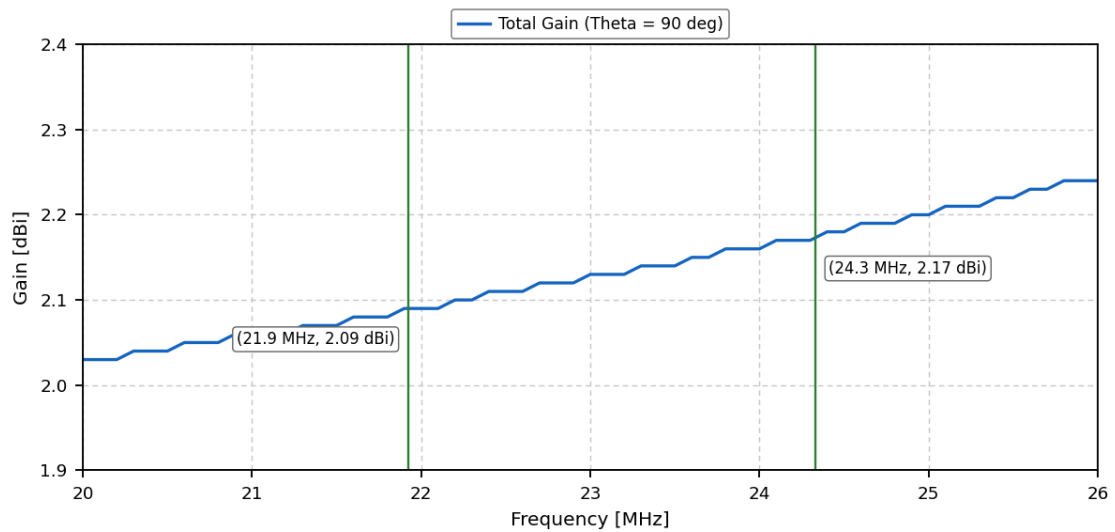


Reflection coefficient Magnitude (Zref = 75 Ohm) - Prob31

Therefore, the bandwidth is $Bw = 24,3 \text{ MHz} - 21,9 \text{ MHz} = 2,4 \text{ MHz}$ (10,5 % relative to 23 MHz).

How does the antenna gain vary over this bandwidth?

Far field



Total Gain (Theta = 90 deg; Phi = 0 deg) - Prob31

The gain increases smoothly and almost linearly across the band, from 2,09 dBi at f_1 to 2,17 dBi at f_2 ($\Delta \approx 0,08$ dB). As the frequency increases, the dipole becomes electrically longer and therefore slightly more directive.

4.2. $\lambda/4$ Monopole

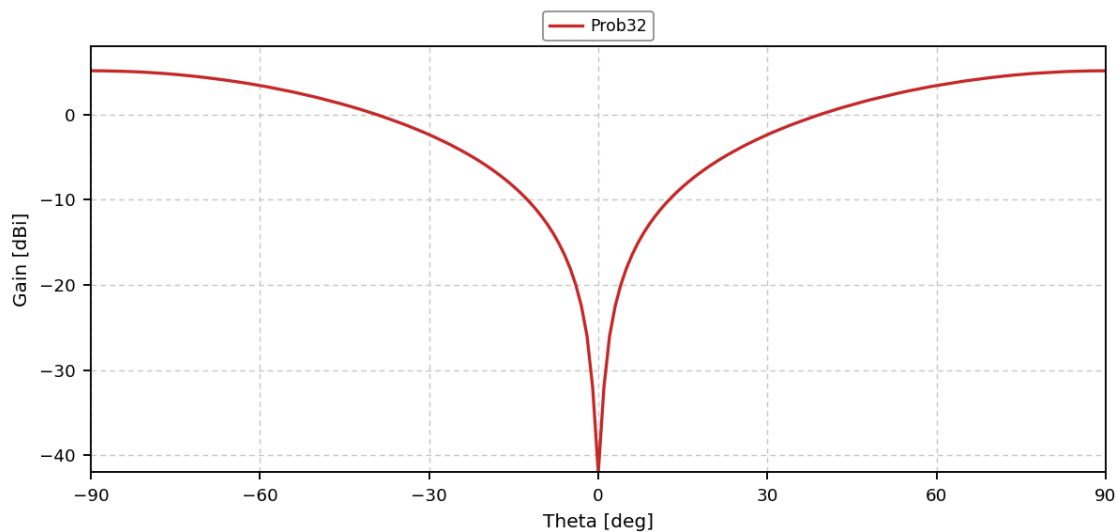
A monopole is designed above a perfect infinite conducting plane by adjusting its height until the input impedance is real:

- Resonant height: $h = 3,0417 \text{ m} = 0,2332\lambda$.
- Input impedance: $3.5958\text{E}+01 + j1.2935\text{E}-06 \Omega$, practically half that of the resonant dipole ($71,8/2 = 35,9 \Omega$), as predicted by image theory.
- Gain: 5,14 dBi, approximately 3 dB higher than the dipole because all the power is radiated into the upper half-space.

→ Radiation pattern in the vertical cut:

Cartesian

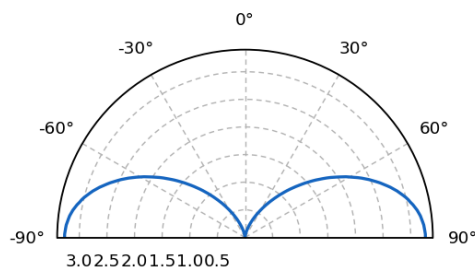
Far field



Total Gain [dBi] (Frequency = 23 MHz; Phi = 0 deg) - Prob32

Polar

Far field



Total Gain (Frequency = 23 MHz; Phi = 0 deg) - Prob32

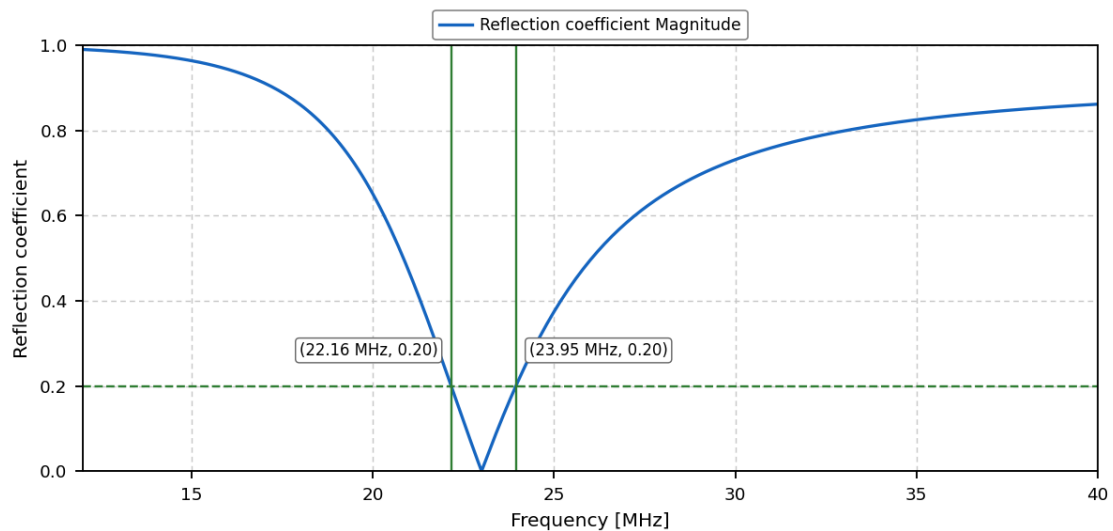
- The -3 dB beamwidth: the maximum is on the horizon ($\theta = \pm 90^\circ$), and the -3 dB points are at $\theta \approx \pm 50,4^\circ$, so the separation between the two points in the vertical cut is

100,8° (equivalent to 39,6° measured from the maximum toward the zenith).

Determine the bandwidth of this monopole if the standing-wave ratio must be below 1,5, using a reference impedance of 36 Ω.

SWR = 1,5 corresponds to $|\Gamma| = (SWR - 1)/(SWR + 1) = 0,5/2,5 = 0,2$:

Reflection coefficient



Reflection coefficient Magnitude (Zref = 36 Ohm) - Prob32

Therefore, the bandwidth is $Bw = 23,95 \text{ MHz} - 22,16 \text{ MHz} = 1,78 \text{ MHz}$ (7,8 %). The fractional bandwidth is smaller than that of the dipole in the previous section, largely because the SWR requirement is now more stringent (1,5 instead of 1,7).

4.3. Monopole Height Reduction

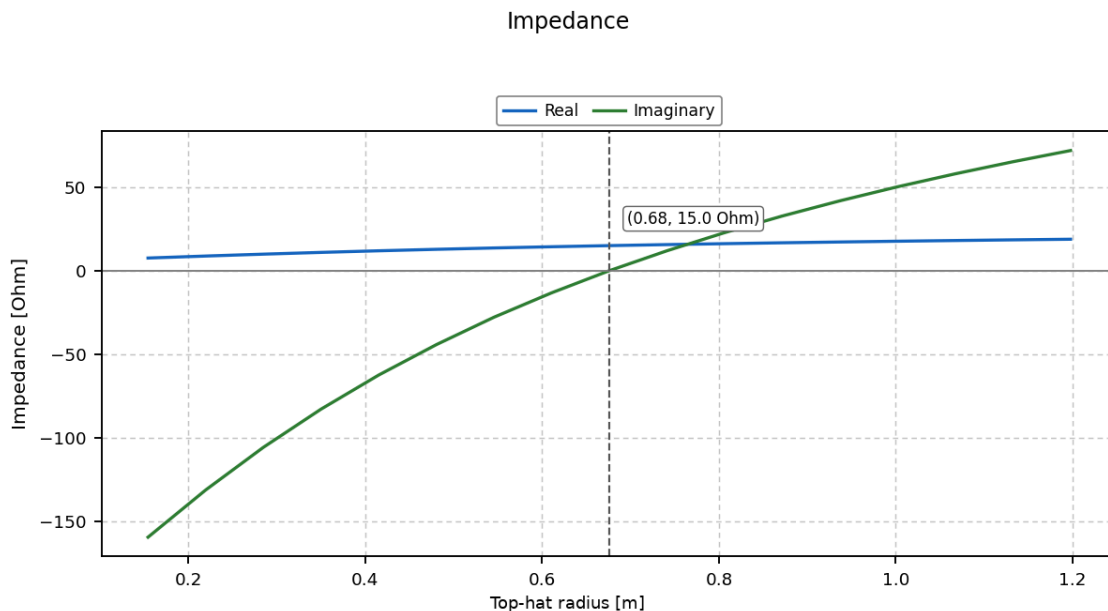
When a monopole is shortened below $\lambda/4$, its radiation resistance decreases rapidly and its reactance becomes strongly capacitive. The next three sections use three classical techniques to retune antennas with height $l \approx 0,1\lambda$: capacitive end loading, bending the wire into an inverted-L antenna, and the inverted-F antenna (IFA).

4.3.1. Prob33A → Shortening a Capacitively Loaded Monopole

The capacitive load is implemented as a top hat consisting of 8 radial wires at the upper end of the monopole. The design procedure is as follows: after fixing the height, the top-hat radius is swept until the reactance is zero; the height is then readjusted so that the resulting resistance has the required value, $Z_{in} = 15 \Omega$. The final design has $h = 1,3354 \text{ m}$ ($0,1024\lambda$) and top-hat radius $R = 0,676 \text{ m}$ ($0,0518\lambda$).

State (for $Z_{in} = 15 \Omega$):

■ Using the Parameter Sweep:



Impedance (Ant_Alt = 1.3354 m; Frequency = 23 MHz)

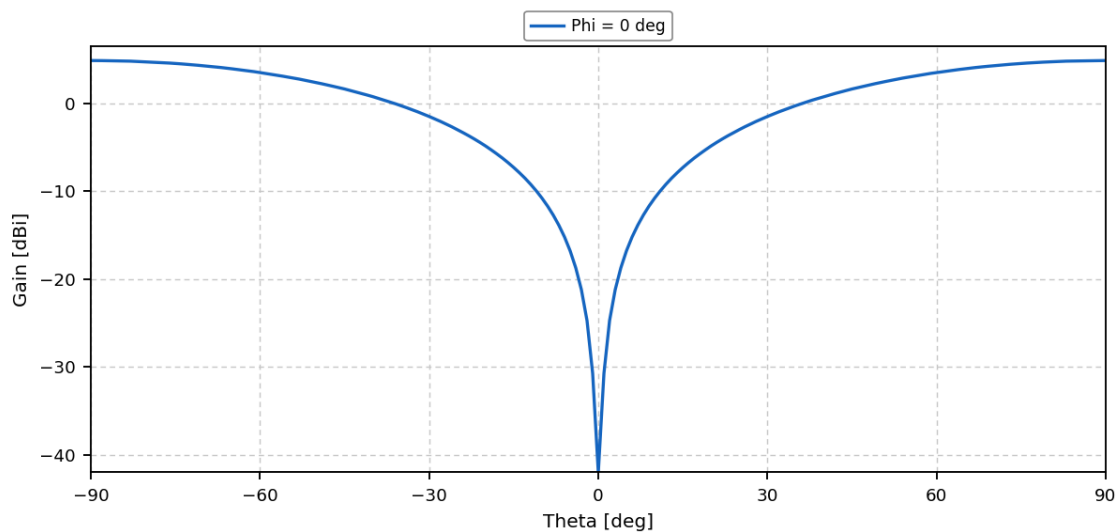
The imaginary part of the impedance crosses zero precisely at $R = 0,676 \text{ m}$, where the real part is $15,0 \Omega$.

- Input impedance: $1.5018\text{E}+01 - j3.8572\text{E}-05 \Omega$.
- Gain: 4,89 dBi.

→ Radiation pattern:

Cartesian

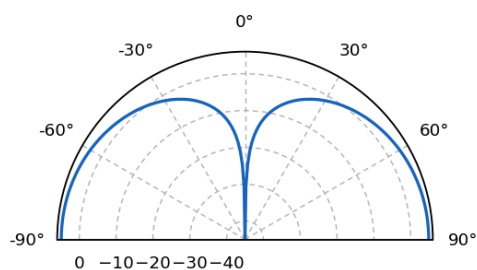
Far field



Total Gain [dBi] (Frequency = 23 MHz; $\Phi = 0^\circ$) - Prob33A

Polar

Far field

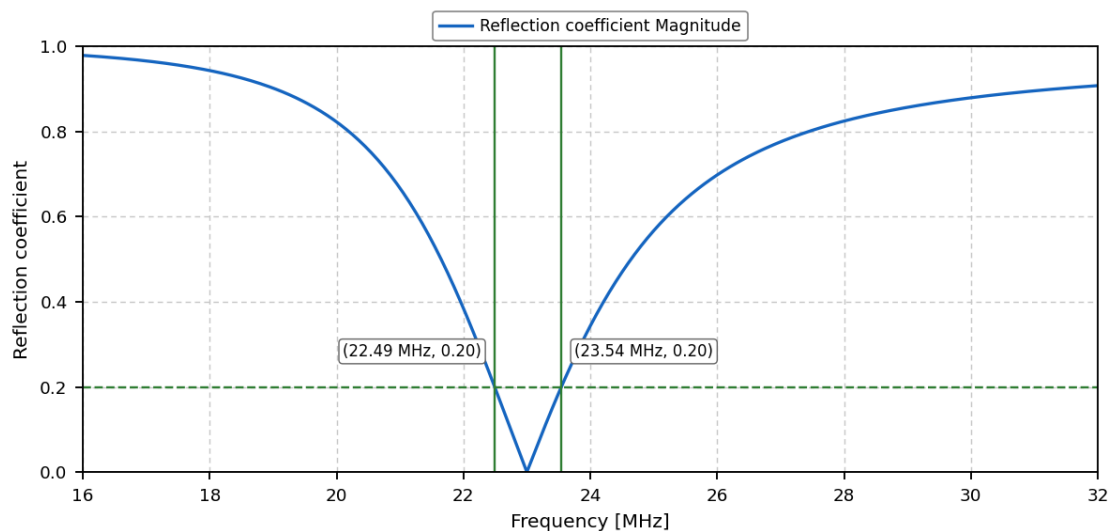


Total Gain [dBi] (Frequency = 23 MHz) - Prob33A

- The -3 dB beamwidth: the -3 dB points are at $\theta \approx \pm 46^\circ$, corresponding to a 93° separation in the vertical cut.

Determine the bandwidth if the standing-wave ratio must be below 1,5, with a reference impedance of 15 Ω . $SWR = 1,5 \Rightarrow |\Gamma| = 0,2$:

Reflection coefficient



Reflection coefficient Magnitude (Zref = 15 Ohm) - Prob33A

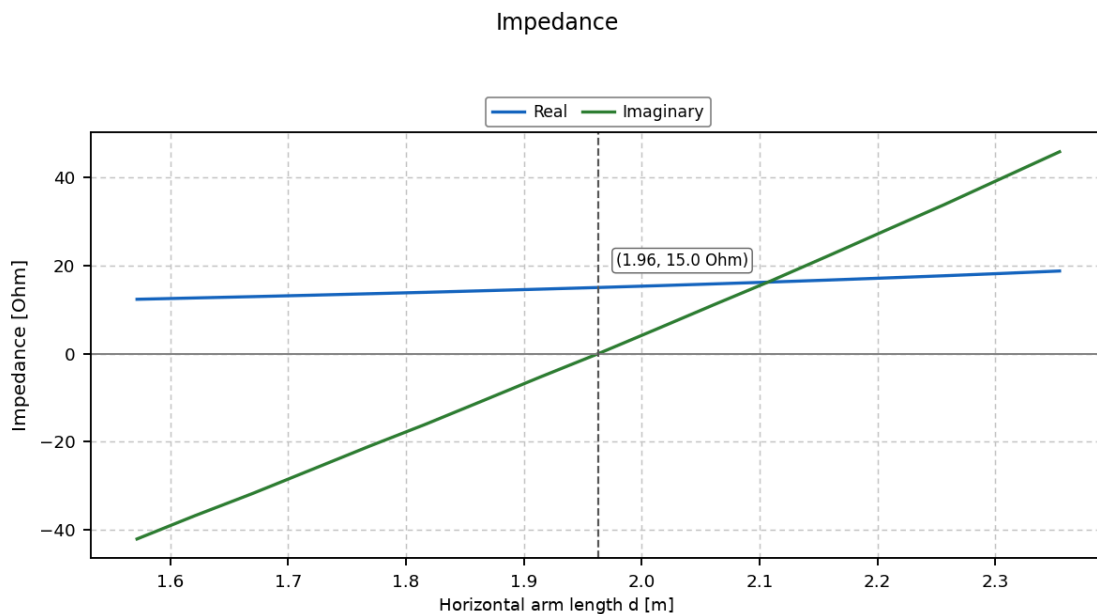
Therefore, $Bw = 23,54 - 22,49 = 1,05$ MHz (4,6 %). The pattern is practically the same as that of the $\lambda/4$ monopole (omnidirectional with a null at the zenith): the currents on the radial wires are horizontal and opposite in pairs, so their radiation cancels, while the vertical current radiates and is now nearly uniform because of the capacitive loading. The gain is only 0,25 dB lower than that of the full-size monopole, but the bandwidth is reduced to little more than half: the shortened antenna stores more reactive energy and therefore has a higher quality factor Q.

4.3.2. Prob33B → Inverted-L Antenna Shortening

The capacitive top hat is replaced by a single horizontal wire, producing an inverted-L antenna. With height $h = 1,2833 \text{ m}$ ($0,0984\lambda$), the length d of the horizontal arm is swept until the reactance is zero. This occurs at $d = 1,9631 \text{ m}$ ($0,1505\lambda$), a value that also provides the required resistance.

State (for $Z_{in} = 15 \Omega$):

■ Using the Parameter Sweep:



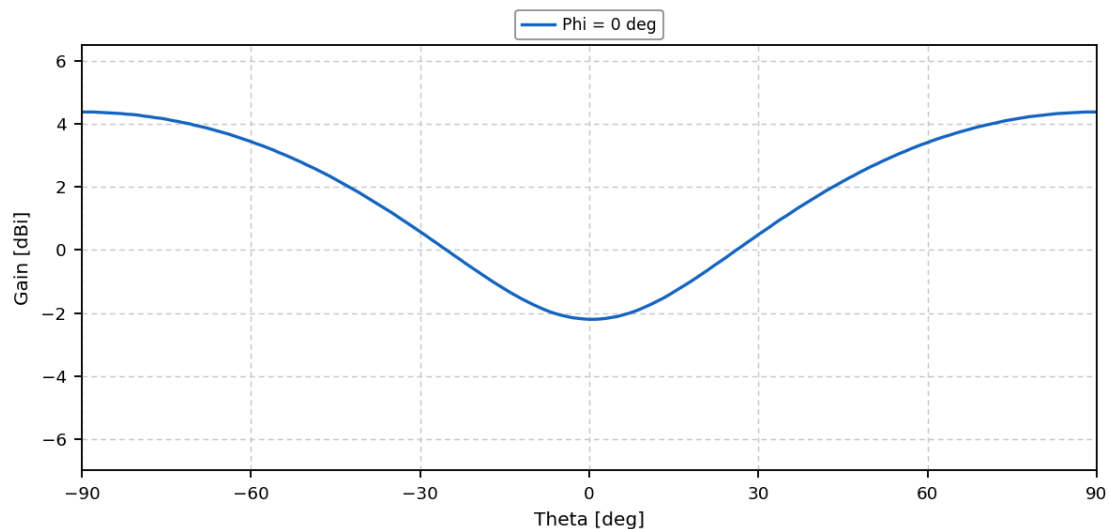
Impedance (Ant_Alt = 1.2833 m; Frequency = 23 MHz)

- Input impedance: $1.5026\text{E}+01 - j7.1722\text{E}-05 \Omega$.
- Gain: 4,38 dBi.

→ Radiation pattern:

Cartesian

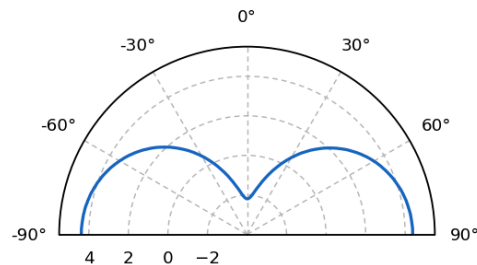
Far field



Total Gain [dBi] (Frequency = 23 MHz; Phi = 0 deg) - Prob33B

Polar

Far field

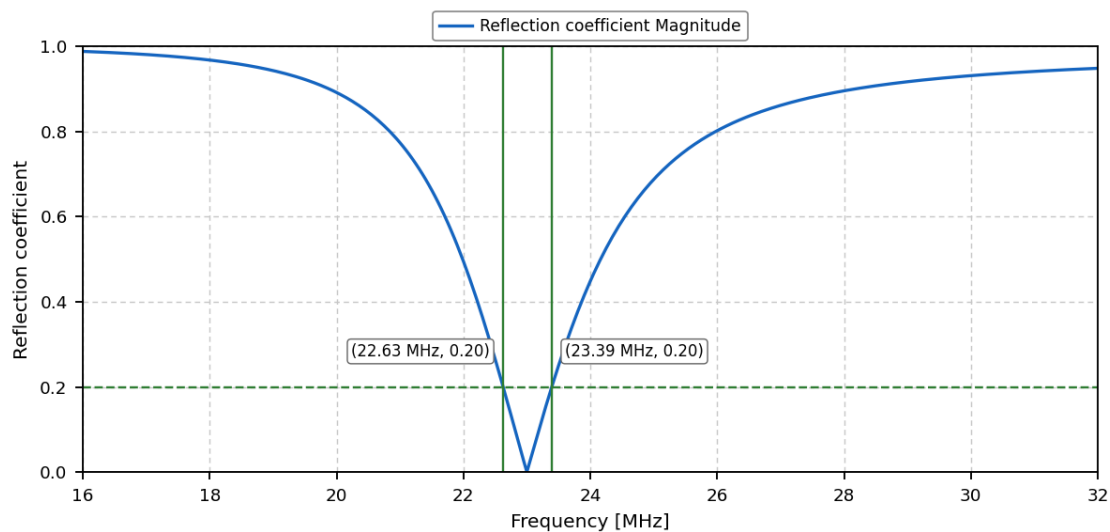


Total Gain [dBi] (Frequency = 23 MHz) - Prob33B

- The -3 dB beamwidth: the -3 dB points are at $\theta = -36,0^\circ$ and $+37,5^\circ$, corresponding to a 73° separation.

Determine the bandwidth if the standing-wave ratio must be below 1,5, with a reference impedance of 15Ω . $SWR = 1,5 \Rightarrow |\Gamma| = 0,2$:

Reflection coefficient



Reflection coefficient Magnitude ($Z_{ref} = 15 \Omega$) - Prob33B

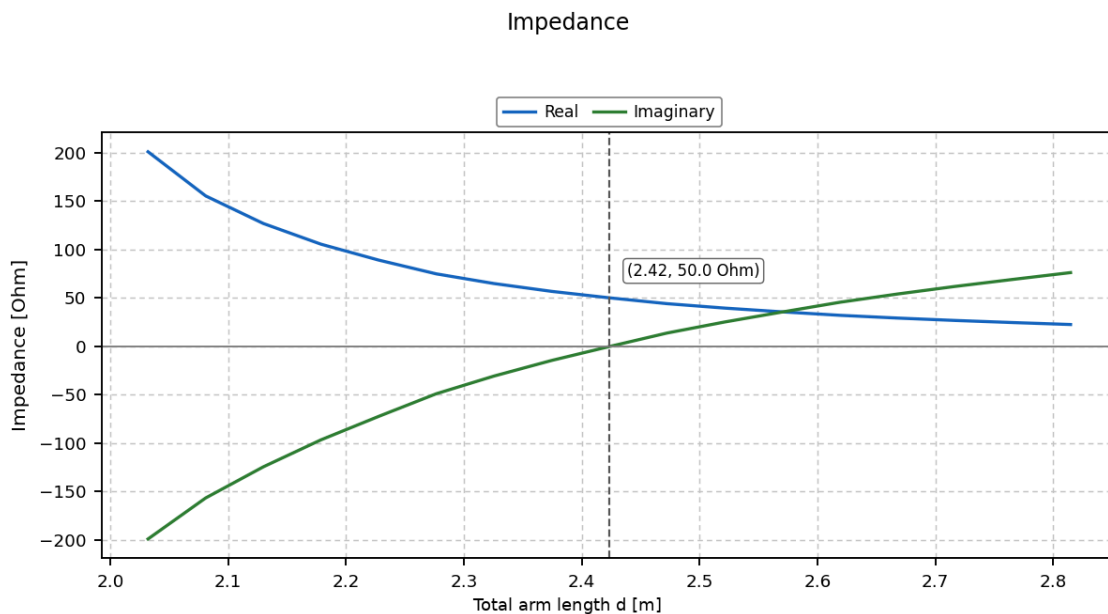
Therefore, $Bw = 23,39 - 22,63 = 0,76 \text{ MHz}$ (3,3 %). Unlike the top hat, the single horizontal arm does radiate: its current does not cancel and adds a horizontally polarized component at the zenith, so the null at $\theta = 0^\circ$ is filled in (the minimum is about 6 dB below the maximum), the pattern becomes narrower, and the gain decreases slightly (4,38 dBi). The bandwidth is reduced again: the equivalent capacitance of the wire is lower than that of the top hat and the vertical current is less uniform, so the Q of the inverted-L antenna is higher.

4.3.3. Prob33C → IFA Antenna

Neither the loaded monopole nor the inverted-L antenna can provide $Z_{in} = 50 \Omega$: for $l \approx 0,1\lambda$, their resistance is limited to approximately 15Ω . The IFA solves this problem by adding a short circuit to ground at the end of the arm and moving the feed a distance s away from the short. The short-to-feed section acts as an impedance transformer that increases the observed resistance. With $h = 0,1\lambda = 1,3043 \text{ m}$, s (which sets the resistance level) and the total arm length d (which sets the resonance) are adjusted simultaneously. The final design for minimum reflection coefficient relative to 50Ω has $s = 0,2806 \text{ m}$ ($0,0215\lambda$) and $d = 2,4234 \text{ m}$ ($0,1858\lambda$).

State (for $Z_{in} = 50 \Omega$):

■ Using the Parameter Sweep:



Impedance (Ant_Alt = 1.3043 m; Ant_s = 0.2806 m; Frequency = 23 MHz)

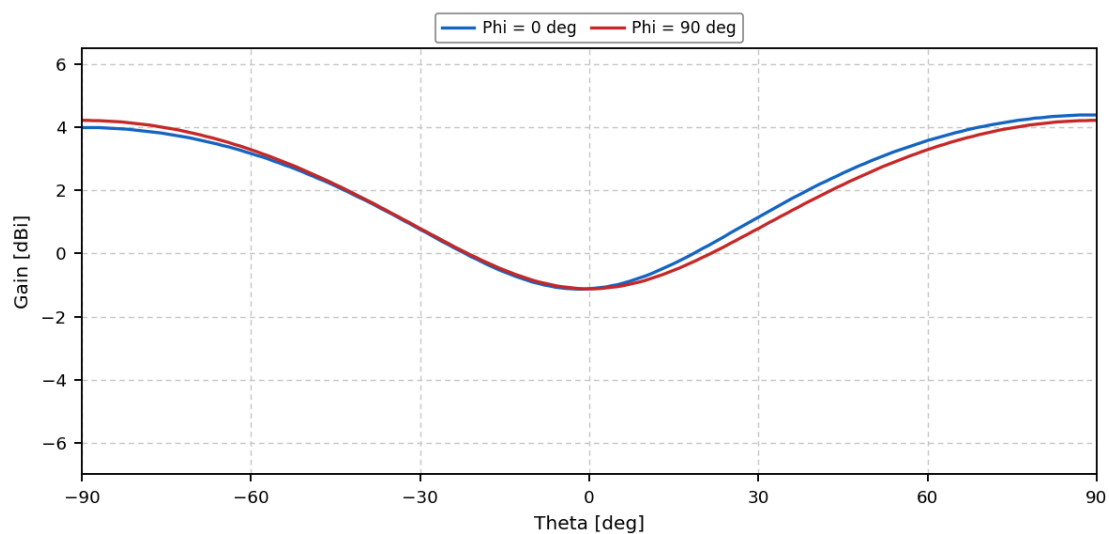
- Input impedance: $5.0039\text{E}+01 - j3.2838\text{E}-01 \Omega$ ($|\Gamma| = 0,003$ at 23 MHz, minimum reflection coefficient).

- Gain: 4,39 dBi.

→ Radiation pattern ($\varphi = 0^\circ$ and $\varphi = 90^\circ$ cuts):

Cartesian

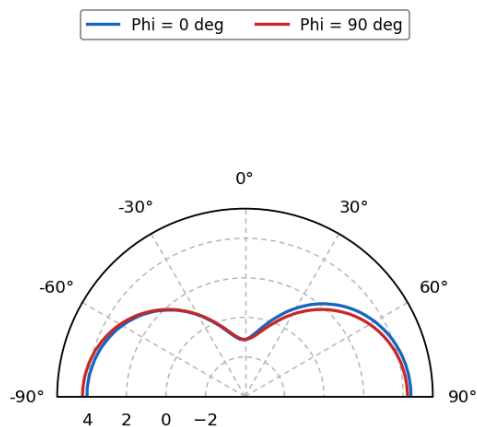
Far field



Total Gain [dBi] (Frequency = 23 MHz; Phi = 0 deg) - Prob33C

Polar

Far field

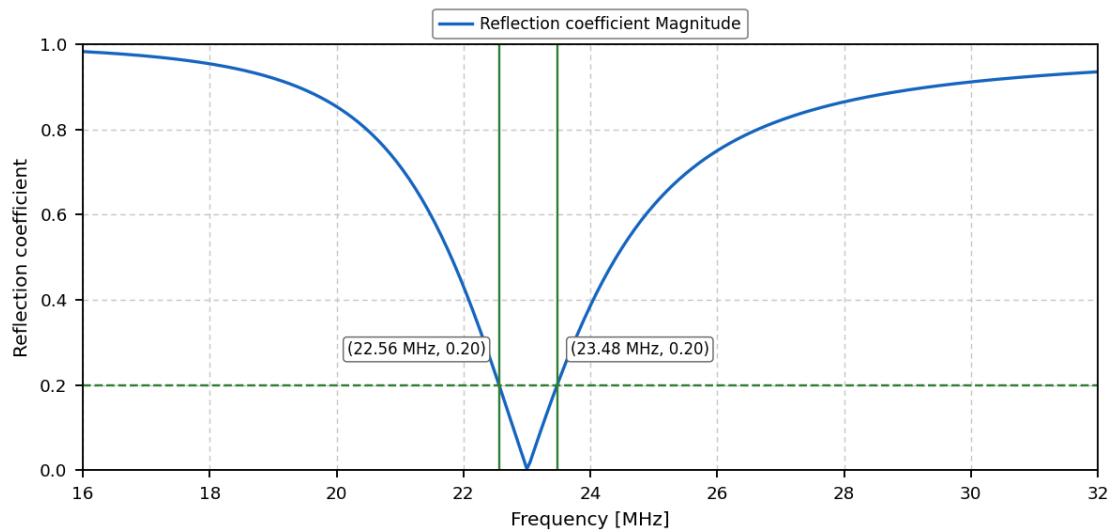


Total Gain [dBi] (Frequency = 23 MHz) - Prob33C

- The -3 dB beamwidth: the -3 dB points are at $\theta = -36,0^\circ$ and $+32,4^\circ$ in the $\varphi = 0^\circ$ cut, corresponding to a 68° separation.

Determine the bandwidth if the standing-wave ratio must be below 1,5 (50 Ω reference). $SWR = 1,5 \Rightarrow |\Gamma| = 0,2$:

Reflection coefficient



Reflection coefficient Magnitude (Zref = 50 Ohm) - Prob33C

Therefore, $Bw = 23,48 - 22,56 = 0,91$ MHz (4,0 %). The $\phi = 0^\circ$ cut shows a slight asymmetry, with the maximum shifted toward the arm side (+x), whereas the $\phi = 90^\circ$ cut is symmetric, as expected from the geometry. The pattern remains quasi-omnidirectional and monopole-like, with the zenith partially filled by radiation from the horizontal arm.

4.4. Loop

Two circular wire loops located in the $z = 0$ plane and fed at $\phi = 0^\circ$ are analyzed: one electrically small loop with length $l = 0,1\lambda$ (radius 0,208 m; wire diameter $l/100 = 13,0$ mm, as specified in the laboratory instructions), and another with length $l = \lambda$ (radius 2,076 m; wire diameter $\lambda/207 = 63,0$ mm, corresponding to the ID number). The table summarizes their parameters:

Length	Input Impedance (Ω)	Gain (dBi)	-3 dB Beamwidth
$0,1\lambda$	$2.0887E-02 + j1.3723E+02$	1,730	92°
λ	$1.1720E+02 - j9.6871E+01$	3,440	134°

The impedance of the small loop agrees with elementary-loop theory: $R = 20\pi^2 \cdot (l/\lambda)^4 = 0,0197 \Omega$ with a high inductive reactance ($+j137 \Omega$). It is an antenna with very low radiation resistance and is difficult to match. In contrast, the one-wavelength loop has values characteristic of a resonant antenna ($R \approx 117 \Omega$) with a moderate capacitive reactance.

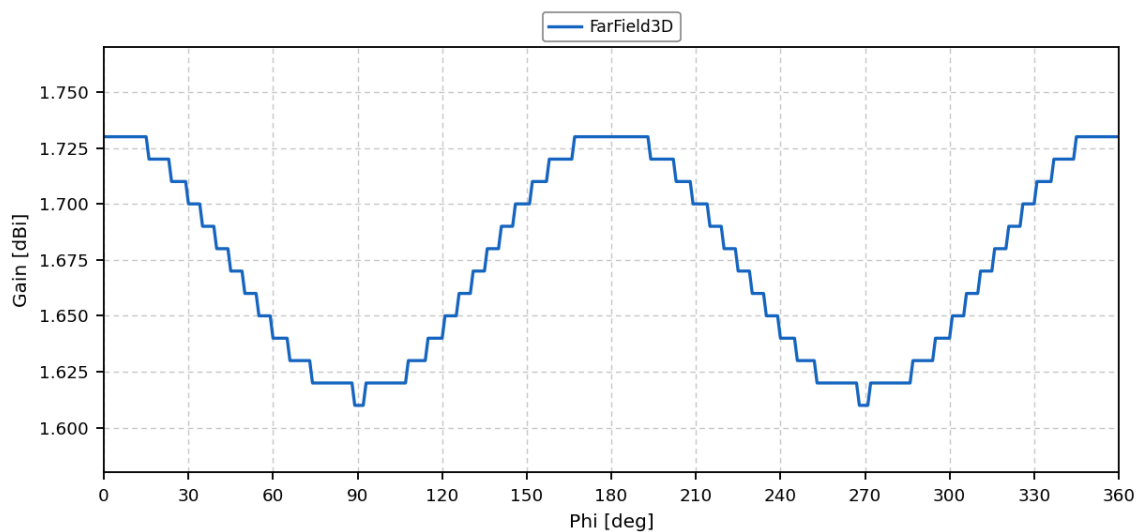
4.4.1. Prob34A

Loop with length $l = 0,1\lambda$.

Maximum direction: $\theta = 90^\circ$ (in the loop plane). The antenna behaves as a z -directed magnetic dipole, with a null along the axis.

→ Radiation pattern in the loop plane (XY plane):

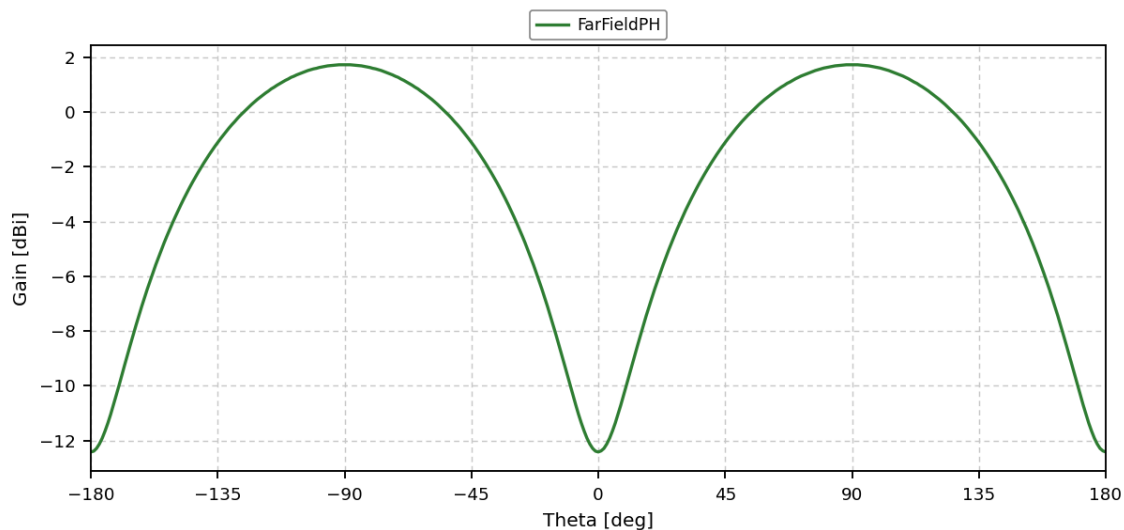
Far field



Total Gain [dBi] (Frequency = 23 MHz; Theta = 90 deg) - Prob34A

→ Radiation pattern in the vertical XZ plane ($\phi = 0^\circ/180^\circ$):

Far field



Total Gain [dBi] (Frequency = 23 MHz; plano XZ, Phi = 0/180 deg) - Prob34A

The XZ plane is selected as the vertical plane because it contains both the direction of maximum radiation and the feed point. It shows the characteristic figure-eight pattern of the magnetic dipole (HPBW = 92°) and, at the same time, any possible asymmetry introduced by the source.

The radiation pattern in the plane containing the loop is not exactly symmetric: the gains at $\phi = 0^\circ$ (1,730 dBi), 90° (1,610 dBi), and 180° (1,730 dBi) are not equal. Because $l \ll \lambda$, the current is almost uniform along the wire and the pattern is nearly circular, but the feed breaks this uniformity and makes the circle slightly elliptical ($\approx 0,12$ dB). Only symmetry about the x axis, which contains the source, is preserved.

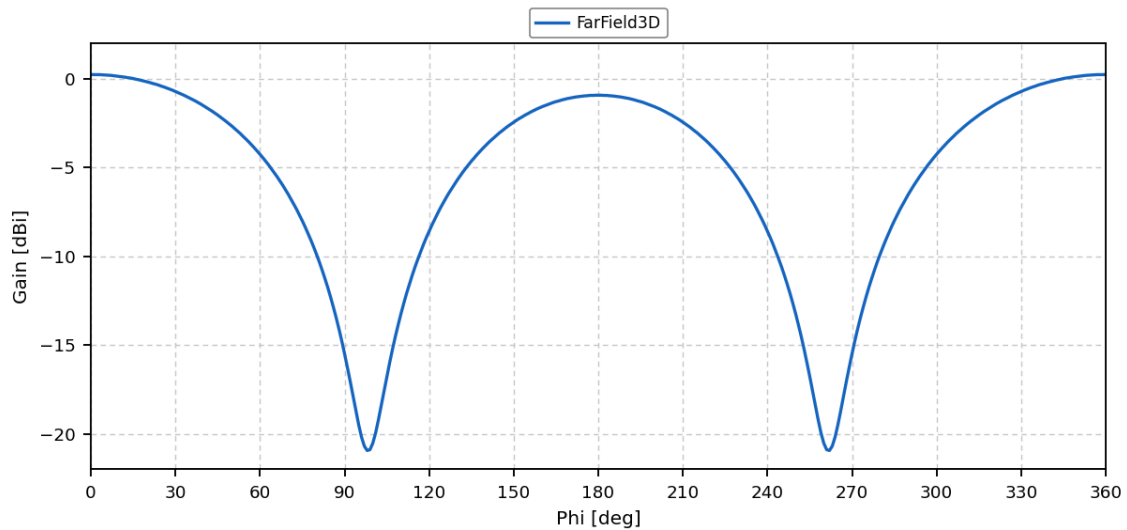
4.4.2. Prob34B

Loop with length $l = \lambda$.

Maximum direction: $\theta = 0^\circ, 180^\circ$ (loop axis).

→ Radiation pattern in the loop plane (XY plane):

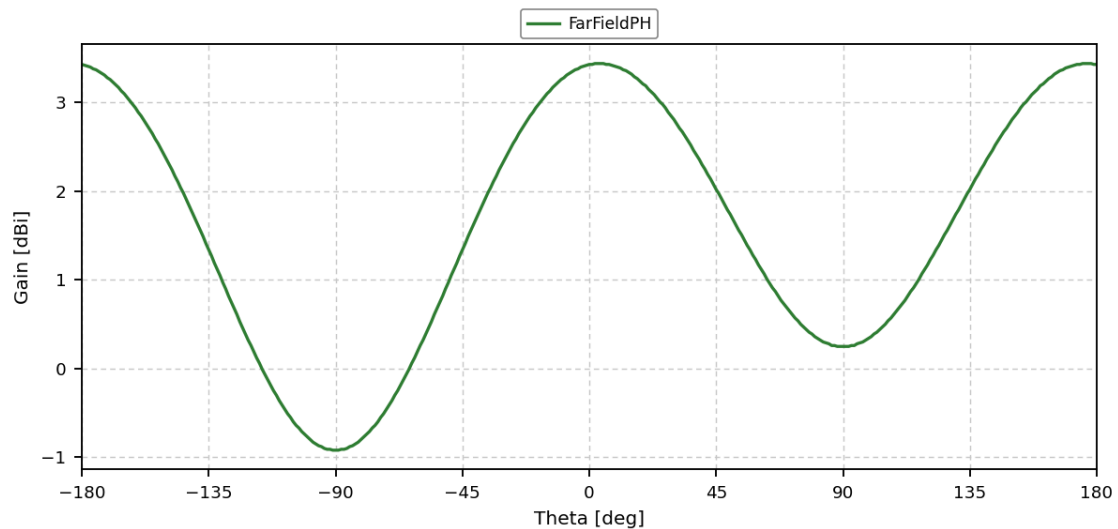
Far field



Total Gain [dBi] (Frequency = 23 MHz; Theta = 90 deg) - Prob34B

→ Radiation pattern in the vertical XZ plane ($\phi = 0^\circ/180^\circ$):

Far field



Total Gain [dBi] (Frequency = 23 MHz; plano XZ, Phi = 0/180 deg) - Prob34B

The vertical XZ plane is retained for the same reason as in the previous case: it contains the two directions of maximum radiation ($\pm z$) and the feed. It shows the two main lobes along the loop axis (3.44 dBi, HPBW = 134°), analogous to those of two parallel in-phase dipoles separated by approximately λ/π .

The radiation pattern in the plane containing the loop is also asymmetric, and much more strongly so in this case: the gains at $\phi = 0^\circ$ (0.25 dBi), 90° (-15.50 dBi), and 180° (-0.92 dBi) differ greatly. The reason is the standing current wave established when $l = \lambda$: current maxima occur at the feed and at the diametrically opposite point, while nulls occur at one quarter of the perimeter. Consequently, different parts of the loop radiate with very different amplitudes and phases, producing deep radiation minima at $\phi = \pm 90^\circ$.